4.2 Exercises

Skill Building

In Problems 1–8, solve each equation for x.

1. $3x + 5 = 20$

2. $2(x - 4) = 10$

3. $7 - x = 3x + 1$

4. $x/3 + 2 = 6$

5. $5x - 9 = 2x + 6$

6. $0.4x = 1.6$

7. $6 = 2(3 - x)$

8. $9x + 4 = 4x - 11$

In Problems 9–16, graph each line on a coordinate grid.

9. $y = 2x - 1$

10. $y = -x + 3$

11. $y = (1/2)x$

12. $x = 4$

13. $y = -2$

14. $2x + y = 4$

15. $y = 3x + 2$

16. $x - y = 1$

In Problems 17–24, plot each point given in polar coordinates.

17. $(3, 45°)$

18. $(2, 120°)$

19. $(-1, 30°)$

20. $(4, 270°)$

21. $(2.5, 210°)$

22. $(-3, 300°)$

23. $(1, 0°)$

24. $(5, 150°)$

In Problems 25–32, find the distance between the two points.

25. $(1, 2)$ and $(4, 6)$

26. $(0, 0)$ and $(5, 12)$

27. $(-2, 3)$ and $(1, -1)$

28. $(6, 6)$ and $(6, -2)$

29. $(-3, -4)$ and $(0, 0)$

30. $(2, 7)$ and $(8, -1)$

31. $(1.5, 2)$ and $(4.5, 6)$

32. $(-1, -1)$ and $(2, 3)$

Applications

33. A ladder leans against a wall. Its foot is at $(0, 0)$ and its top touches the wall at $(0, 12)$. The wall is 5 feet from the foot of the ladder. How long is the ladder?

34. A drone starts at the origin and flies to the point with polar coordinates $(40, 60°)$. Find its rectangular coordinates, to the nearest tenth of a meter.